

&" NuclearCraft Plutonium Age

Complete Player Guide

Crafting Recipes · Custom Items · Machines · Radiation · Combat · Boss

PaperMC 1.21+ · Plugin Version 1.0 · Phases 1–10

All values configurable via YAML config files in the plugin folder

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Material Substitution Reference

NuclearCraft custom items cannot be directly used in vanilla crafting grids because Minecraft's recipe system does not understand Custom Model Data. Instead, each custom item uses a vanilla material as a stand-in. When you place that material in a crafting grid, the plugin validates the real item via its CustomModelData tag before accepting the recipe.

Crafting Grid Substitutions

Vanilla Material (place in grid)	Represents Custom Item	CMD
Echo Shard	Refined Plutonium Ingot	1103
Yellow Wool	Industrial Fabric	1315
Magma Cream	Radioactive Core	1101
Prismarine Crystals	Raw Plutonium Fragment	1102
Heart of the Sea	Irradiated Heart	1106
Pink Petals	Healing Petal	1105
Wheat Seeds	Mutated Seed	1104
Nether Star	Titan Core OR Reactor Heart (context-dependent)	1107/1109
Amethyst Shard	Titan Fragment OR Mutated Crystal	1108/1111

Block & Item Base Materials

Vanilla Base	Represents	CMD
Blast Furnace	Nuclear Smelter block	1201
Smithing Table	Nuclear Forge block	1401
Diamond Pickaxe	Radiation Drill	1108
Leather Helmet/Chest/Legs/Boots	Hazmat Suit pieces	1306–1309
Netherite Helmet/Chest/Legs/Boots	Plutonium Armor pieces	3001–3004
Netherite Sword/Pickaxe/Axe/Shovel/Hoe	Plutonium Tools	2001–2005
Honey Bottle	Radiation Antidote	1301
Glass Bottle	Radiation Serum	1302
Paper	Ancient Reactor Blueprint	1110

&" CMD = CustomModelData — the unique number that makes each item visually distinct in-game.

&" Always use the correct custom item (obtained via /nc give commands or gameplay) — placing a plain vanilla Echo Shard without CMD will not work.

All Custom Items — Materials & Components

Raw Materials

Raw Plutonium Fragment

Base: Prismarine Crystals | CMD: 1102

Mined from Plutonium Ore (Y: -64 to -58). 1–4 per ore block. Low radiation while in inventory. Feed into Nuclear Smelter to refine.

Refined Plutonium Ingot

Base: Echo Shard | CMD: 1103

Output of the Nuclear Smelter. Primary crafting material for all plutonium weapons, tools, and armor. Also used as repair material in an Anvil (+300 durability per ingot).

Radioactive Core

Base: Magma Cream | CMD: 1101

Dropped by Irradiated Zombies. Two uses: fuel for the Nuclear Forge (2,000 energy per Core) and ingredient in some recipes. High radiation source — keep in lead-lined storage.

Irradiated Heart

Base: Heart of the Sea | CMD: 1106

Dropped by Irradiated Zombies (rare). Required ingredient for MK-3 and MK-4 Nuclear Forge upgrades. Also required for the Titan summoning ritual (1 Heart). Emits strong radiation.

Industrial Fabric

Base: Yellow Wool | CMD: 1315

Radiation-hardened woven material. Used to craft all four Hazmat Suit pieces. Also the repair material for Hazmat armor in an Anvil (+200 durability per piece).

Mutated Seed

Base: Wheat Seeds | CMD: 1104

Dropped by Irradiated Zombies (5–50% chance by level). Plant on Radioactive Farmland to grow Mutated Healing Plants. Cannot be planted on normal farmland.

Healing Petal

Base: Pink Petals | CMD: 1105

Harvested from fully grown Mutated Healing Plants (1–3 per harvest). Primary ingredient for Radiation Antidote and Radiation Serum. Right-clicking also gives Regen II for 5s and removes 25 radiation.

All Custom Items — Equipment, Machines & Consumables

Machines & Tools

Nuclear Smelter

Base: Blast Furnace | CMD: 1201

Converts Raw Plutonium Fragments to Refined Ingots. Right-click to open GUI. Requires fuel (Coal, Charcoal, Coal Block, Blaze Rod, Lava Bucket). Emits 8 radiation per 5 seconds within 3 blocks while active.

Nuclear Forge

Base: Smithing Table | CMD: 1401

Upgrades plutonium and hazmat gear to MK-I through MK-IV. Uses Radioactive Cores as energy fuel (2,000 energy each). Right-click to open GUI. Overloading causes a radiation burst.

Radiation Drill

Base: Diamond Pickaxe | CMD: 1108

The ONLY tool that safely mines Plutonium Ore. Mining without it gives a +25 radiation burst and drops nothing. Diamond+ mining speed (×1.2 modifier). 1,800 durability. Also has a 10% chance to find extra Radioactive Debris.

Consumables & Cures

Radiation Antidote

Base: Honey Bottle | CMD: 1301

Right-click to consume. Clears ALL radiation instantly. Removes infection status and all radiation debuffs. Does NOT grant immunity — you can be re-infected immediately. Crafted with Healing Petals + Honey Bottle.

Radiation Serum

Base: Glass Bottle | CMD: 1302

Right-click to consume. Instantly clears ALL radiation. Removes all debuffs. ALSO grants 10 minutes of full radiation immunity — you cannot be irradiated during this window. Expensive to craft.

Boss Items (Titan Drops)

Titan Core

Base: Nether Star | CMD: 1107

Crystallised essence of the Plutonium Titan. 4 Titan Cores are consumed when performing the summoning ritual to call the Titan. EXTREME radiation hazard — keep in lead-lined storage.

Titan Fragment

Base: Amethyst Shard | CMD: 1108

Shard of the Titan's reactor plating. Dropped on kill. Used in high-tier crafting (Phase 11+). Emits extreme radiation.

Reactor Heart

Base: Nether Star | CMD: 1109

The beating core of the Titan's reactor. Rare drop (10% chance). Required for MK-3 and MK-4 Nuclear Forge upgrades. Critical radiation hazard.

Mutated Crystal

Base: Amethyst Shard | CMD: 1111

Crystallised radioactive energy grown inside the Titan. 8% drop chance. Used in future high-tier crafting.

Ancient Reactor Blueprint

Base: Paper | CMD: 1110

Very rare Titan drop (5% chance). Unlocks advanced reactor crafting in Phase 11. No current gameplay use — save for future updates.

Crafting Recipes — Machines & Special Items

All recipes crafted at a standard Crafting Table unless otherwise noted.
Proxy materials in parentheses are what you actually hold; the plugin checks for the custom item.

Nuclear Smelter

D = Diamond O = Obsidian F = Furnace

Diamond	Obsidian	Diamond
Obsidian	Furnace	Obsidian
Diamond	Obsidian	Diamond

Result: 1× Nuclear Smelter block

Nuclear Forge

I = Refined Plutonium Ingot (Echo Shard proxy) D = Ancient Debris O = Obsidian T = Smithing Table

Obsidian	Anc.Debris	Obsidian
Ref.Ingot	Sm.Table	Ref.Ingot
Obsidian	Anc.Debris	Obsidian

Result: 1× Nuclear Forge block |
Ref.Ingot = Echo Shard in grid

Lead-Lined Storage Crate

L = Lead (Iron Ingot proxy) W = Wood Planks C = Chest

Iron Ingot	Iron Ingot	Iron Ingot
Iron Ingot	Chest	Iron Ingot
Iron Ingot	Iron Ingot	Iron Ingot

Result: 1× Lead-Lined Storage Crate |
Stores fragments WITHOUT emitting
radioactive particles

Plutonium Block

Ref.Ingot	Ref.Ingot	Ref.Ingot
Ref.Ingot	Ref.Ingot	Ref.Ingot
Ref.Ingot	Ref.Ingot	Ref.Ingot

Result: 1× Plutonium Block (compact storage) | Ref.Ingot = Echo Shard proxy | Emits more radiation than single ingots — handle with full Plutonium armor

Crafting Recipes — Plutonium Tools

P = Echo Shard (= Refined Plutonium Ingot) S = Stick All tools use Netherite base.

Radiation Drill

R = Radioactive Core (Magma Cream proxy) D = Diamond

R-Core	Diamond	R-Core
Diamond	Diamond	Diamond
	Stick	

Result: 1× Radiation Drill | Only tool that safely mines Plutonium Ore | ×1.2 durability

Plutonium Sword

	Ref.Ingot	
Plutonium Sword	Ref.Ingot	
	Stick	

Plutonium Pickaxe

DMG: +10 +10 radiation on hit +5 rad vs Stage 3+ players	Speed: 1.7/s Ref.Ingot	Dur: 2,500 +20 on crit Ref.Ingot
Plutonium Pickaxe	Stick	
Plutonium Hoe	Stick	

Durability: 2,800
10% chance to find Radioactive Debris
Requires Radiation Drill to mine
Plutonium Ore

Ref.Ingot	Ref.Ingot	
Ref.Ingot	Stick	Plutonium Axe
	Stick	

DMG: +11 15% shockwave (3-block radius, +15 rad) AoE radiation burst on shockwave	Speed: 1.0/s Ref.Ingot	Dur: 2,600
	Stick	Plutonium Shovel
	Stick	

Durability: 2,500
15% chance to convert soil to
Radioactive Farmland on dig

Ref.Ingot	Ref.Ingot	
	Stick	
	Stick	

Result: Plutonium Hoe | Durability: 2,000 | Fast crop tilling

Crafting Recipes — Armor

P = Echo Shard (= Refined Plutonium Ingot) · W = Yellow Wool (= Industrial Fabric)

Plutonium Armor uses Netherite as base; Hazmat Suit uses Leather.

Plutonium Armor (P = Echo Shard)

Full set: Prot III + Unbreaking III on each piece | Full set grants environmental radiation IMMUNITY + Speed II + Fire Resistance

P	P	P	P		P
P	Plutonium Helmet	P	P	P	Plutonium Chestplate
P	P	P			
HELMET — 4 armor · 2 tough · 450 dur			CHESTPLATE — 8 armor · 2 tough · 640 dur		
-15% radiation per piece			-15% radiation per piece		
P	Plutonium Leggings	P	P		Plutonium Boots
P		P	P		

Full set bonus: 80% radiation reduction total | Based on Leather armor | Repair with Industrial Fabric

590 dur	-15% radiation per piece	W	dur	-15% radiation · Feather Falling II	W
			No radiation from ground contact		
W		W	W	W	W
Hazmat Helmet					Hazmat Chestplate
			W	W	W
W	W	W			
HELMET — 2 armor · 363 dur · -20% rad			CHESTPLATE — 5 armor · 529 dur · -30% rad		
W		W	W		W
Hazmat Leggings					Hazmat Boots
W		W	W		W

LEGGINGS — 4 armor · 496 dur · -20% rad

BOOTS — 2 armor · 430 dur · -10% rad

Crafting Recipes — Cures & Arrows

Radiation Antidote (Shapeless Recipe)

Shapeless = ingredients can be in any slots in the crafting grid.

Healing Petal

Healing Petal

Honey Bottle

!

Radiation Antidote
x1

Effect: Instantly clears ALL radiation, infection, and debuffs. Does NOT grant immunity.

Radiation Serum (Shaped Recipe)

H = Healing Petal R = Radioactive Core (Magma Cream proxy) G = Gold Nugget
A = Golden Apple B = Glass Bottle

Healing Petal

R-Core

Healing Petal

Gold Nugget

Golden Apple

Gold Nugget

Healing Petal

Glass Bottle

Healing Petal

Result: 1× Radiation Serum | Effect:
Clears ALL radiation + grants 10

Plutonium Arrow (Shaped Recipe)

R-Core = Radioactive Core (Magma Cream proxy)
P = Refined Plutonium Ingot (Echo Shard proxy) A = Vanilla Arrow F = Feather

Ref.Ingot

Arrow

Feather

Result: 4× Plutonium Arrows
On hit: +25 radiation · Poison I (4 seconds) · Clawing (5 seconds)
Critical hit (full bow charge): +20 bonus radiation

Smelter Recipe: Raw Fragment ! Refined Ingot

INPUT

Raw PlutFrag

!

Refined Ingot

Time: 15 seconds · Requires fuel · Machine must reach 500°C

Radiation System — Stages & Sources

Radiation is measured in points (0–1000). It accumulates from exposure to radioactive materials, zombies, and machines. As your level rises, debuffs worsen and direct damage begins.

Radiation Stages

Stage	Name	Range	Effects
Stage 0	Healthy	0–99	None
Stage 1	Minor Exposure	100–249	Weakness I · Nausea
Stage 2	Moderate Exposure	250–499	Weakness I · Slowness I · Nausea
Stage 3	Severe Exposure	500–749	Weakness II · Slowness II · Hunger I · 0.5 HP damage per cycle
Stage 4	Critical Poisoning	750–1000	Weakness III · Slowness III · Hunger II · 1.0 HP damage per cycle · Contagious!

Natural Progression & Decay

Progression (while infected)	Stage 1: +5 pts every 60s · Stage 2: +10 · Stage 3: +15 · Stage 4: +20
Natural Decay (when clean)	Requires 10 minutes with NO new radiation !' removes 2 pts per 60 seconds

All Radiation Sources

Source	Amount	Trigger
Plutonium Ore (within 1 block)	+5 pts	Every 5 seconds
Plutonium Ore (within 2 blocks)	+2 pts	Every 5 seconds
Plutonium Ore (within 3 blocks)	+1 pt	Every 5 seconds
Mining ore WITHOUT Radiation Drill	+25 burst	Per attempt (5s cooldown)
Raw Plutonium Fragment in inventory	+1 pt	Every 30 seconds
Plutonium Block (within 1 block)	+10 pts	Every 5 seconds
Plutonium Block (within 2 blocks)	+5 pts	Every 5 seconds
Plutonium Block (within 3 blocks)	+2 pts	Every 5 seconds
Radioactive Debris (within 5 blocks)	+0.5/s	Continuous passive
Nuclear Smelter active (3-block radius)	+8 pts	Every 5 seconds
Radiation Cloud (zombie death)	+5/s	10 seconds in 3-block radius
Irradiated Zombie Lv.1 hit	+10 pts	Per melee hit
Irradiated Zombie Lv.2 hit	+20 pts	Per melee hit
Irradiated Zombie Lv.3 hit	+35 pts	Per melee hit
Alpha Zombie Lv.4 hit	+50 pts	Per melee hit
Irradiated Zombie aura (nearby)	+2 pts	Every 5 seconds within range
Titan Boss — Radiation Aura	+3/s	10-block radius, all phases
Titan Boss — Titan Slam	+15 pts	Per slam hit
Titan Boss — Radiation Wave	+40 pts	20-block radius burst
Titan Boss — Reactor Overload	+70 pts	18-block radius burst
Titan Boss — Energy Beam	+60 pts	Targeted player hit
Titan Boss — Nuclear Catastrophe	+80 pts	30-block radius (Phase 4 only)
MK-IV Forge Overload	+150 pts	8-block radius burst
Toxic Bloom (farming hazard)	+8 pts	Every 2 seconds, 4-block radius

Radiation System — Protection & Contagion

Radiation Cures

Item	How to Use	Effect
Healing Petal (direct)	Right-click to consume	Removes 25 radiation · Regen II for 5 seconds
Radiation Antidote	Right-click to consume	Clears ALL radiation, infection & debuffs · No immunity
Radiation Serum	Right-click to consume	Clears ALL radiation & debuffs · 10 min full immunity

Armor Radiation Protection

Armor reduces all incoming radiation exposure by a flat multiplier per piece.

Armor Type / Situation	Radiation Reduction	Notes
Any single armor piece (generic)	-15% per piece	Up to -60% for full 4-piece set
Hazmat Helmet	-20%	Leather base
Hazmat Chestplate	-30%	Best single piece for rad protection
Hazmat Leggings	-20%	Leather base
Hazmat Boots	-10%	No ground radiation special
Full Hazmat Suit (all 4 pieces)	-80% total	Best accessible radiation shield
Plutonium Armor — full set (4/4)	IMMUNE	Zero environmental radiation
Plutonium Boots (special)	No ground radiation	Negates radioactive farmland etc.
Near Plutonium Ore (hazmat)	×0.20 multiplier	80% of ore radiation blocked
Near Plutonium Ore (plut. armor)	×0.00 multiplier	Complete immunity to ore zones

Contagion — Radiation Spread Between Players

Irradiated players at Stage 2+ can spread radiation to others through proximity, physical contact, or shared vehicles.

Spread Method	Condition	Amount	Chance
Proximity (within 3 blocks)	Infected is Stage 2	+25 pts	5% per 5s
Proximity (within 3 blocks)	Infected is Stage 3	+25 pts	15% per 5s
Proximity (within 3 blocks)	Infected is Stage 4	+25 pts	25% per 5s
Physical contact (melee hit)	Any stage	+50 pts	35% per hit
Shared vehicle (boat/minecart)	Any stage	+25 pts	20% per 15s

Immunity Details

Source	Duration	Stacks?
Radiation Serum (item)	10 minutes	No — resets timer
Plutonium Armor (environmental)	Permanent while worn	Only blocks environmental sources
Base game immunity (built-in)	30s default / up to 5 min	60s cooldown between refreshes

Particle & Sound Indicators

Radiation stages are visually indicated by coloured particles around the player:

Stage 1 (Minor)	Bright green particles [RGB 57, 255, 20]
Stage 2 (Moderate)	Yellow particles [RGB 255, 220, 0]
Stage 3 (Severe)	Orange particles [RGB 255, 100, 0]
Stage 4 (Critical)	Dark red particles [RGB 200, 0, 0]

A Geiger counter sound effect also plays based on your current radiation level.

Nuclear Smelter — Machine Guide

Placement & Usage

1. Craft a Nuclear Smelter (Diamond + Obsidian + Furnace pattern — see Page 6)
2. Place it like any block in the world
3. Right-click the Smelter to open its GUI
4. Insert Raw Plutonium Fragments in the INPUT slot
5. Insert fuel in the FUEL slot
6. The machine heats up, then begins processing
7. Collect Refined Plutonium Ingots from the OUTPUT slot

GUI Slot Layout

Input Slot	Place Raw Plutonium Fragments here
Fuel Slot	Insert Coal, Charcoal, Coal Block, Blaze Rod, or Lava Bucket
Output Slot	Collect your Refined Plutonium Ingots here
Progress Bar	Shows current processing progress and temperature

Fuel Values

Fuel Item	Units	Approx. Processing Time
Coal	100 units	~20 seconds of active processing
Charcoal	80 units	~16 seconds
Coal Block	900 units	~3 minutes (most efficient per slot)
Blaze Rod	120 units	~24 seconds
Lava Bucket	1,000 units	~3 min 20 sec !• BEST fuel per bucket

Temperature System

The Smelter must reach a minimum temperature before processing begins.

Parameter	Value
Ambient (idle) temperature	20°C
Minimum temperature to process	500°C (must be reached before any processing)
Maximum safe temperature	1,500°C
Heating rate (fueled, idle)	+5°C per tick cycle
Active heating rate (processing)	+2.5°C per tick cycle
Cooling rate (idle, no fuel)	-1°C per tick cycle
Overheat duration	10,000 ms (10 seconds) before cooling begins
Overheat cooling rate	-2°C per tick (twice as fast as normal)
Restart threshold	80% of 500°C = 400°C (no cold start needed)

Radiation Hazard

The active Smelter emits passive radiation to all players within 3 blocks.

Passive radiation amount	+8 radiation points per check
Check interval	Every 5 seconds (100 ticks)
Affected radius	3 blocks around the machine block

&" Build the Smelter in an enclosed room with no direct access. Full Hazmat Suit reduces this to ~1.6 pts/check.

&" Lava Bucket gives the best processing time per inventory slot — carry several for long sessions.

Nuclear Forge — Upgrade Machine Guide

What It Does

The Nuclear Forge upgrades Plutonium tools, weapons, and Hazmat armor pieces from their base state up to MK-I, then to MK-II, MK-III, and finally MK-IV. Each upgrade tier adds percentage bonuses to damage, speed, armor, and durability.

How to Use

1. Craft and place a Nuclear Forge block (see Page 6 for recipe)
2. Right-click to open the upgrade GUI
3. Place your item in the INPUT slot
4. Insert Radioactive Cores in the FUEL slot (each Core = 2,000 energy)
5. Insert Refined Plutonium Ingots in the MATERIAL slot
6. For MK-3 and MK-4: also insert Irradiated Hearts in the HEART slot
7. Click UPGRADE — the process takes several seconds
8. On success: your item is upgraded | On failure: item stays the same (MK-4 has 5% downgrade risk)

Upgrade Tiers — Full Details

Tier	Ingots	Cores	Hearts	Energy	Success%	Time	Bonus
MK-1	2	1	0	500	100%	3 sec	+5% dmg, speed, armor, dur
MK-2	4	2	0	1,200	90%	5 sec	+10% all stats
MK-3	8	4	1	2,500	75%	8 sec	+20% all stats
MK-4	16	8	2	5,000	50%	12 sec	+35% all stats · 5% downgrade risk on fail
&" Ingots = Refined Plutonium Ingots (Echo Shard proxy) · Cores = Radioactive Cores · Hearts = Irradiated Hearts							
&" MK-4 50% success: on failure, item keeps current tier. 5% additional chance of downgrade to MK-3.							

MK-IV Special — Radiation Aura

Wearing any MK-IV upgraded item activates a passive Radiation Aura around the player.

Aura Parameter	Value
Aura radius	3 blocks around player
Radiation to hostile mobs per tick	+25 radiation
Radiation to enemy players (PvP)	+15 radiation per tick
Aura tick rate	Every 2 seconds (40 ticks)
Affects team-mates?	No (only hostile mobs and enemies)

Overload System

If too much energy is loaded (>10,000 units) the Forge OVERLOADS.

Overload Parameter	Value
Max energy before overload	10,000 units
Energy per Radioactive Core	2,000 units
Energy decay (idle)	-0.5 units per tick (auto-bleeds excess)
Overload shutdown duration	200 ticks (10 seconds)
Radiation burst on overload	+150 radiation to all players in 8-block radius
Block destroyed on overload?	No (safe — forge survives)
&" Never insert more than 5 Radioactive Cores at once — that reaches the max energy limit.	

World — Ore, Mining & Farming

Plutonium Ore Spawning

Parameter	Value
Y-level range	Y: -64 to -58 (deepest deepslate layer only)
Dimension	Overworld only
Spawn chance per chunk	60% of newly-generated chunks get a vein
Vein size	1–2 blocks per vein (very small — rare resource)
Valid host blocks	Deepslate, Stone, Cobbled Deepslate, Tuff
Discovery alert range	5 blocks — chat notification when nearby

Mining Rules

Plutonium Ore is dangerous. You MUST use the Radiation Drill to safely extract it.

Scenario	Outcome
Mine with any tool EXCEPT Radiation Drill	+25 radiation burst · Ore does NOT drop · 5s cooldown
Mine with Radiation Drill (no Fortune)	1 Raw Plutonium Fragment · 3–7 XP · No radiation burst
Mine with Radiation Drill + Fortune I	1–2 Raw Plutonium Fragments
Mine with Radiation Drill + Fortune II	1–3 Raw Plutonium Fragments
Mine with Radiation Drill + Fortune III	1–4 Raw Plutonium Fragments
Mine with Radiation Drill + Silk Touch	1× Plutonium Ore block (no fragments)

Radioactive Farming — How to Farm

Step-by-step farming process:

1. Get Mutated Seeds from Irradiated Zombie drops (all levels)
2. Create Radioactive Farmland: dig normal dirt/grass with a Plutonium Shovel (15% chance per block)
3. Right-click to plant Mutated Seeds on Radioactive Farmland ONLY — other farmland rejects them
4. Wait for the plant to grow through 5 stages (0 = seed planted, 4 = fully bloomed)
5. Right-click to harvest at Stage 4 — you get Healing Petals and bonus seeds
6. The crop replants itself (some seeds drop)

Farming Stats & Drops

Parameter	Value
Growth stages	0 (seed) !' 1 !' 2 !' 3 !' 4 (mature blossom)
Radioactive Farmland growth bonus	+50% extra random-tick growth chance
Bone meal compatible?	Yes — vanilla bone meal works normally
Healing Petals per harvest	1–3 Petals
Mutated Seeds per harvest	0–2 Seeds (self-sustaining supply)
XP per harvest	2–5 XP
Toxic Bloom chance	1% — fully grown crop turns dangerous instead of dropping
Farmland passive radiation	+1 pt every 2 seconds to nearby players

Toxic Bloom — Farming Hazard

Rarely (1%), a fully-grown Mutated Healing Plant mutates into a Toxic Bloom instead of dropping normally. It immediately begins irradiating the area.

Toxic Bloom Parameter	Value
Radiation radius	4 blocks
Radiation amount	+8 points per 2 seconds
How to remove	Break it with any tool — it drops nothing

&" Always harvest Mutated Healing Plants as soon as they reach Stage 4 to minimise Toxic Bloom risk.

Irradiated Zombies

Spawning Rules

60% of all naturally-spawning zombies become Irradiated Zombies. Baby zombies are excluded.
Eligible spawn reasons: Natural, Jockey, Mount, Reinforcements, Village Defence, Chunk Gen.

Level Stats

Level (Spawn %)	HP	DMG	Speed	KB Res	Rad/Hit	XP
Lv.1 — Standard (80%)	35 HP	5 DMG	×1.15	0.15	+10 pts	10 XP
Lv.2 — Enhanced (15%)	45 HP	7 DMG	×1.20	0.20	+20 pts	20 XP
Lv.3 — Powerful (4%)	60 HP	9 DMG	×1.25	0.30	+35 pts	40 XP
Lv.4 — Alpha (1%)	80 HP	12 DMG	×1.30	0.50	+50 pts	100 XP

&" Level 4 Alpha: displays green Bossbar "&" Alpha Irradiated Zombie &"", Glowing effect, visible from 20 blocks. Treat as a mini-boss.

Loot Drops

Item	Lv.1	Lv.2	Lv.3	Lv.4 Alpha
Rotten Flesh	0–3 always	0–3 always	0–3 always	0–3 always
Radioactive Core (Magma Cream)	15%	25%	40%	100% (ALWAYS)
Mutated Seed (Wheat Seeds)	5%	10%	20%	50%
Irradiated Heart (Heart of Sea)	1%	3%	7%	25%

Radiation Cloud (On Zombie Death)

Spawn chance per death	20% chance
Cloud radius	3 blocks
Duration	10 seconds
Radiation inside cloud	+5 points per second

&" Avoid standing near freshly-killed Irradiated Zombies — even at low levels the cloud is dangerous.

Radiation Surge Night Event

Trigger chance per night	5%
Effect on radiation	ALL radiation damage doubled for the full night
Effect on loot	ALL zombie drops doubled
Player alert	Bossbar shown: &" RADIATION SURGE ACTIVE &"
Chat broadcast	"&" RADIATION SURGE! A wave of radioactive energy sweeps the world!"

Zombie Advancements & XP Rewards

First Exposure	+50 XP — first radiation hit from any zombie
Mutant Hunter	+100 XP — kill first Irradiated Zombie
Core Collector	+100 XP — collect first Radioactive Core
Alpha Slayer	+500 XP — kill first Level 4 Alpha Zombie

Plutonium Titan — Boss Guide

Summoning Ritual

You need: 4× Titan Cores in your inventory + 1× Irradiated Heart in your hand.

ALTAR LAYOUT:

Center block: Crying Obsidian

4 Corner blocks: Obsidian (placed 2 blocks diagonally from the center on the same Y level)

STEPS:

1. Build the altar (Crying Obsidian center, Obsidian corners)
2. Have 4× Titan Cores anywhere in your inventory
3. Right-click the center Crying Obsidian while holding an Irradiated Heart
4. All 4 Titan Cores are consumed immediately from your inventory
5. A 15-second summoning animation plays
6. The Plutonium Titan spawns at the altar center

COOLDOWN: 60 minutes between summons. Only one Titan can exist at a time.

Health Scaling (by nearby player count)

Players Nearby	Titan HP
1–3 players	5,000 HP
4–6 players	6,500 HP
7–10 players	8,500 HP
10+ players	10,000 HP

Phase System

Phase (HP%)	Speed Mult.	DMG Mult.	Active Abilities
Phase 1 (100%–75%)	×1.00	×1.00	Radiation Aura · Titan Slam
Phase 2 (75%–50%)	×1.10	×1.20	Phase 1 + Radiation Wave · Mutant Summoning
Phase 3 (50%–25%)	×1.25	×1.50	Phase 2 + Reactor Overload · Energy Beam
Phase 4 (25%–0%)	×1.50	×2.00	Phase 3 + Nuclear Catastrophe · Final Frenzy

All Titan Abilities

Ability	Phases	Radius	Damage/Radiation	Cooldown
Radiation Aura (passive)	All	10 blocks	+3 rad/sec continuously	None
Titan Slam	1, 2, 3, 4	5 blocks	12 DMG + 15 rad + knockback	10 sec
Radiation Wave	2, 3, 4	20 blocks	+40 radiation burst (ring)	15 sec
Mutant Summon	2, 3, 4	—	3 Lv.1 + 1 Lv.4 Alpha zombie	25 sec
Reactor Overload	3, 4	18 blocks	+70 radiation burst	40 sec
Energy Beam	3, 4	Targeted	25 DMG + 60 radiation	20 sec
Nuclear Catastrophe	4 only	30 blocks	+80 rad · sets all HP to 1	90 sec
Final Frenzy	4 only	—	+50% attack speed forever	One-time

&" Nuclear Catastrophe: 10-second charge-up before the blast — watch for the animation and retreat!

&" Mutant Summon spawns a Level 4 Alpha every 25 seconds from Phase 2 — clear them quickly!

Titan Rewards (Per Contributing Player)

Reward	Amount / Chance	Condition
Server XP	5,000 XP total (split by contribution)	Any contribution
Titan Fragments	1–8+ depending on contribution %	Any contribution
Refined Plutonium Ingots	Scales with contribution	Minimum 5% contribution

Reactor Heart (Nether Star)	10% chance	Qualifying players
Ancient Reactor Blueprint	5% chance	Qualifying players
Mutated Crystal	8% chance	Qualifying players
Loot Chest at death location	1× chest with Cores, Fragments, Ingots	All participants

Combat System & Admin Commands

Plutonium Weapon Effects

Weapon / Hit Type	Radiation Applied	Additional Effect
Plutonium Sword (base hit)	+10 radiation	None
Sword — Critical hit (falling)	+10 + 5 bonus = 15	Extra radiation on crit
Sword — vs Stage 3+ victim	+10 + 5 bonus = 15	Bonus when victim already infected
Plutonium Axe (base hit)	+0 base radiation	15% chance: 3-block shockwave +15 rad
Axe — Shockwave (15%)	AoE +15 radiation	3-block radius, all targets hit
Plutonium Arrow (hit)	+25 radiation	Poison I (4s) · Glowing (5s)
Arrow — Critical (full charge)	+25 + 20 = 45	Bonus radiation on full pull
Arrow — Headshot bonus	+25 + 10 = 35	Approximated headshot detection

Combo System

Max combo stack	8 hits
Combo reset time	6 seconds after last hit
Bonus radiation per stack	+5 per hit above the first (hit 2 = +5, hit 3 = +10...)
Maximum combo bonus	+35 radiation (capped regardless of hit count)

Weapon Mastery Levels

Level	XP Required	Unlocks / Bonus
Novice	0 XP	Starting level — no bonuses
Experienced	100 XP	+1 XP/sword hit · +10 XP/kill
Veteran	500 XP	+1 XP/axe hit · +10 XP/kill
Elite	1,500 XP	+2 XP/bow hit · +15 XP/kill
Master	4,000 XP	+10% radiation on all weapon hits · +10% arrow radiation · +1.5 block axe shockwave radius

Equipment Repair

Repair Plutonium tools/armor	Anvil + Refined Plutonium Ingot (Echo Shard proxy) !' +300 durability per ingot
Repair Hazmat Suit pieces	Anvil + Industrial Fabric (Yellow Wool proxy) !' +200 durability per piece

Admin / Operator Commands (/nuclearcraft or /nc)

Command	Description
/nc radiation set <player> <0-1000>	Set a player's radiation to a specific value
/nc radiation add <player> <amount>	Add radiation points to a player
/nc radiation clear <player>	Clear all radiation from a player
/nc radiation status [player]	View current radiation stage and points
/nc ore give fragment [amount]	Give Raw Plutonium Fragments
/nc ore give drill	Give a Radiation Drill
/nc equipment give <type>	Give plutonium/hazmat item (sword/axe/pickaxe/shovel/hoel/helmet/chestplate/leggings/boots/hazmat-helmet/hazmat-chestplate/hazmat-leggings/hazmat-boots)
/nc farming give seed [amount]	Give Mutated Seeds
/nc farming give petal [amount]	Give Healing Petals
/nc farming give antidote [amount]	Give Radiation Antidotes
/nc farming give serum [amount]	Give Radiation Serums
/nc smelter give	Give a Nuclear Smelter block

/nc forge give	Give a Nuclear Forge block
/nc titan spawn	Force-spawn the Plutonium Titan at your location
/nc titan kill	Instantly despawn the active Titan
/nc titan phase <1-4>	Force the Titan to a specific combat phase
/nc reload	Reload all YAML config files without restart

All values are configurable in the plugin YAML files inside the server's `/plugins/NuclearCraft/` folder.

